

Alex Coker

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Education

University of New Mexico

Expected Spring 2027

B.S. in Mechanical Engineering | GPA: 3.45/4.0

- Relevant coursework: dynamics, control systems, numerical methods
- Actively preparing PhD applications in robotics (target: Fall 2027 enrollment)

Research Experience

Learning and Control Lab — University of New Mexico

Oct. 2025 – Present

Undergraduate Research Assistant | Advisor: Prof. Leilei Cui

Control Barrier Function Safety Filter on the Unitree Go2

- Implemented Aaron Ames-style CBF as a real-time safety filter wrapping the Go2's RL locomotion policy; filter minimally edits velocity commands to enforce hard keep-out zones while the learned policy handles low-level tracking
- Prototyped as a closed-form single-integrator CBF, validated in MuJoCo with barrier $h(t) \geq 0$ and robot stability confirmed throughout
- Ported filter to C++ and deployed on physical Go2 (Unitree SDK2 + LibTorch on Jetson Orin); ran live MuJoCo digital twin in parallel for side-by-side real/sim comparison
- Extended perception front-end from camera-based to LiDAR-based keep-out zone detection, keeping CBF filter logic unchanged downstream

GitHub: [pseudower1/go2-cbf](https://github.com/pseudower1/go2-cbf)

MPPI-CBF Integration for Safe Quadruped Navigation

- Integrated MPPI sampling-based trajectory planning with CBF safety filtering to enable autonomous point-to-point navigation on the Go2
- MPPI planner generates obstacle-avoiding trajectories; CBF layer enforces formal clearance guarantees — eliminating keep-out zone violations that occur with plain MPPI
- Validated in simulation across cluttered fields, maze, and gauntlet courses; hardware transfer ongoing

GitHub: [pseudower1/go2-mppi-cbf](https://github.com/pseudower1/go2-mppi-cbf)

Broader Lab Contributions

- Collaborated with PhD students on Go2 handstand project (whole-body RL policy, sim-to-real transfer)
- Conducted literature review of safety-critical control: CBFs, Hamilton-Jacobi reachability analysis, and MPPI

Industry Experience

Sandia National Laboratories

June 2025 – Present

Mechanical Engineering Intern | PV Reliability Group

- Designed and executed experiments characterizing PV connector failure mechanisms under thermomechanical loading across thousands of samples
- Performed four-wire (Kelvin) resistance measurement and controlled electrical testing; analyzed data to identify degradation and failure modes
- Presented results at DOE quarterly program reviews and to 100+ industry experts at the PVQAT solar reliability conference
- Recipient of the Sandia Thunderbird Intern Award (Courageous) for initiative and excellence

Projects

Autonomous Drone Search-and-Sample RL Controller

- Trained a PPO policy (Stable-Baselines3 / gym-pybullet-drones) to fly a simulated quadrotor across unknown terrain, searching for biosignature targets under a finite battery and time budget while avoiding decoy sites
- Benchmarked against lawnmower-sweep, gradient-follower, and random-walk baselines; trained policy detected more targets per episode than all baselines
- Developed hybrid controller (RL search + lawnmower-style homing on detection) that outperformed all single-strategy approaches on collection rate

GitHub: [pseudower1/rl-drone](https://github.com/pseudower1/rl-drone)

CubeSat ADCS — Slew Maneuver & Pointing Budget

- Simulated a reaction-wheel-stabilized 6U CubeSat tracking ground targets during overhead passes; held a 0.1° pointing requirement with ~2.5× margin across three passes
- Extended to inter-satellite tracking scenario (6.95 km/s crossing encounter) and identified the momentum saturation condition that causes the design's margin to run out

Custom Robotic Arm — From-Scratch Control

- Reverse-engineered control scheme, joint mapping, and command interface of a legacy desktop arm with no surviving SDK or documentation
- Built a complete control pipeline from hardware probing through motion sequencing, achieving reliable manipulation motion

Technical Skills

Programming: Python, C++

Robotics & Control: CBF safety filters, MPPI, Hamilton-Jacobi reachability, RL policy deployment, sim-to-real transfer, legged robots (Unitree Go2), Unitree SDK2, LibTorch

Simulation: MuJoCo, Isaac Gym, PyBullet, digital twin development

Hardware & Lab: Jetson Orin, LiDAR, camera-based perception, Kelvin resistance measurement, experimental design, hand tools

Tools & Systems: Git, Linux, Windows, network & computer setup, Claude Code

Honors & Awards

- Sandia Thunderbird Intern Award — Courageous (Sandia National Laboratories, 2025)
- UNM Rainforest Innovations Pitch Competition — Arid Sustainability Award (two-time winner)
- Lobo Hackathon — Second Place
- Design II Final Project — First Place Team Award (mousetrap car competition)

Outreach

- GearMasters Volunteer — mentored K–12 students in hands-on STEM and engineering activities